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Sounding and transmission precoding matrix indication determination using machine learning models

Abstract

Certain aspects of the present disclosure provide techniques for sounding and precoding uplink transmissions using one or more machine learning models. An example method generally includes generating a sounding reference signal (SRS) using an SRS deep neural network (DNN); transmitting, to a network entity, the generated SRS in one or more resource elements (REs); receiving, from the network entity, information identifying a precoding matrix to use for uplink transmissions on a shared channel; precoding uplink transmissions based on the identified precoding matrix; and transmitting, to the network entity, the precoded uplink transmissions on the shared channel.

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References Cited

U.S. PATENT DOCUMENTS

Patent No.	Issued Date	Patentee Name	U.S. Cl.	CPC
2019/0277957	12/2018	Chandrasekhar	N/A	G01S 5/017
2020/0336264	12/2019	Faxér	N/A	H04B 7/0617
2021/0064996	12/2020	Wang et al.	N/A	N/A
2021/0120499	12/2020	Su	N/A	H04W 52/14
2021/0399773	12/2020	Huang	N/A	H04B 17/364
2022/0338189	12/2021	Madadi	N/A	H04B 17/3913

FOREIGN PATENT DOCUMENTS

Patent No.	Application Date	Country	CPC
2020077536	12/2019	WO	N/A
2021081981	12/2020	WO	N/A

OTHER PUBLICATIONS

International Search Report and Written Opinion—PCT/CN2021/093542—ISA/EPO—Oct. 25, 2021. cited by applicant
Vivo: “Study on AI/ML based Air Interface Enhancement in Rel-18”, 3GPP TSG RAN Meeting #91e, RP-210321, 3rd Generation Partnership Project, Mobile Competence Centre, 650, Route Des Lucioles, F-06921 Sophia-Antipolis Cedex, France, vol. TSG RAN, No. Electronic Meeting, Mar. 16, 2021-Mar. 26, 2021, 13 Pages, Mar. 15, 2021, XP051985674, the whole document. cited by applicant

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Background/Summary

CROSS-REFERENCE TO RELATED APPLICATION(S)
(1) This application is a national stage application under 35 U.S.C. 371 of PCT/CN2021/093542, filed May 13, 2021, which is hereby expressly incorporated by reference herein in its entirety as if fully set forth below and for all applicable purposes.
INTRODUCTION
(2) Aspects of the present disclosure relate to wireless communications, and more particularly, to techniques for sounding and transmission precoding matrix indication determination for uplink transmissions using machine learning models.
(3) Wireless communication systems are widely deployed to provide various telecommunication services such as telephony, video, data, messaging, broadcasts, or other similar types of services. These wireless communication systems may employ multiple-access technologies capable of supporting communication with multiple users by sharing available system resources with those users (e.g., bandwidth, transmit power, or other resources). Multiple-access technologies can rely on any of code division, time division, frequency division orthogonal frequency division, single-carrier frequency division, or time division synchronous code division, to name a few. These and other multiple access technologies have been adopted in various telecommunication standards to provide a common protocol that enables different wireless devices to communicate on a municipal, national, regional, and even global level.
(4) Although wireless communication systems have made great technological advancements over many years, challenges still exist. For example, complex and dynamic environments can still attenuate or block signals between wireless transmitters and wireless receivers, undermining various established wireless channel measuring and reporting mechanisms, which are used to manage and optimize the use of finite wireless channel resources. Consequently, there exists a need for further improvements in wireless communications systems to overcome various challenges.
SUMMARY
(5) One aspect provides a method for wireless communication by a user equipment. The method generally includes generating a sounding reference signal (SRS) using an SRS deep neural network (DNN); transmitting, to a network entity, the generated SRS in one or more resource elements (REs); receiving, from the network entity, information identifying a precoding matrix to use for uplink transmissions on a shared channel; precoding uplink transmissions based on the identified precoding matrix; and transmitting, to the network entity, the precoded uplink transmissions on the shared channel.
(6) One aspect provides a method for wireless communication by a network entity. The method generally includes receiving, from a user equipment (UE), a sounding reference signal generated by an SRS deep neural network (DNN); identifying a precoding matrix for the UE to use in uplink transmissions based on a precoding matrix DNN and the received SRS; transmitting, to the UE, information identifying the precoding matrix; and receiving, from the UE, uplink transmissions precoded using the identified precoding matrix.
(7) Other aspects provide: an apparatus operable, configured, or otherwise adapted to perform the aforementioned methods as well as those described elsewhere herein; a non-transitory, computer-readable media comprising instructions that, when executed by one or more processors of an apparatus, cause the apparatus to perform the aforementioned methods as well as those described elsewhere herein; a computer program product embodied on a computer-readable storage medium comprising code for performing the aforementioned methods as well as those described elsewhere herein; and an apparatus comprising means for performing the aforementioned methods as well as those described elsewhere herein. By way of example, an apparatus may comprise a processing

system, a device with a processing system, or processing systems cooperating over one or more networks.
(8) The following description and the appended figures set forth certain features for purposes of illustration.

Description

BRIEF DESCRIPTION OF THE DRAWINGS
(1) The appended figures depict certain features of the various aspects described herein and are not to be considered limiting of the scope of this disclosure.
(2) FIG. **1** is a block diagram conceptually illustrating an example wireless communication network.
(3) FIG. **2** is a block diagram conceptually illustrating aspects of an example a base station and user equipment.
(4) FIGS. **3**A-**3**D depict various example aspects of data structures for a wireless communication network.
(5) FIG. **4** illustrates a call flow diagram of messages that may be exchanged between a UE and a network entity (such as a gNodeB) to precode an uplink transmission based on SRSs and precoding matrices generated or otherwise identified by one or more machine learning models.
(6) FIG. **5** illustrates example operations for wireless communication by a user equipment for sounding and performing uplink transmissions using machine learning models, in accordance with certain aspects of the present disclosure.
(7) FIG. **6** illustrates example operations for wireless communication by a network entity (such as a gNodeB) for configuring a user equipment (UE) to perform uplink transmissions using machine learning models, in accordance with certain aspects of the present disclosure.
(8) FIG. **7** depicts aspects of an example communications device
(9) FIG. **8** depicts aspects of an example communications device.
DETAILED DESCRIPTION
(10) Aspects of the present disclosure provide apparatuses, methods, processing systems, and computer-readable mediums for sounding and transmission precoding matrix indication determination for uplink transmissions using machine learning models.
(11) Precoding schemes are generally used in beamforming transmissions to maximize a received signal strength of the transmission. Generally, the precoding used for transmissions may be determined based on various channel quality measurements. For example, precodings can be determined based on channel condition similarities assumed between an uplink channel and a downlink channel. However, because uplink channel conditions may be different from downlink channel conditions, the assumed channel condition similarities between the uplink and downlink channels may not actually exist. Thus, a precoding for an uplink transmission determined based on downlink channel conditions may not be an appropriate precoding for the uplink transmission.
(12) Aspects of the present disclosure provide techniques for machine learning-based sounding and precoding for uplink transmissions. Generally, signaling used for uplink sounding by a network entity may be generated by a reference signal generation machine learning model, and the network entity can process the received signaling using a transmission precoding identification machine learning model trained to identify a transmission precoding matrix for the UE based on the received signaling. A precoding machine learning model may be used at the UE to interpret the received identification and identify the precoding to be used for uplink transmission to the network entity.
(13) By using machine learning models for sounding and precoding for uplink transmissions, a UE may perform uplink transmissions using precodings that are more appropriate for the uplink channel conditions than a precoding assumed to be appropriate based on assumed channel

condition similarities. By doing so, processing overhead in determining an appropriate precoding for an uplink transmission may be reduced. This may result in more efficient use of wireless communications resources, improved reliability for communications between the network entity and the UE, power savings due to a reduction in the amount of data transmitted and/or received in determining a precoding for uplink transmissions and in performing re-transmission of failed transmissions, and the like.

Introduction to Wireless Communication Networks

(14) FIG. **1** depicts an example of a wireless communications system **100**, in which aspects described herein may be implemented.

(15) Generally, wireless communications system **100** includes base stations (BSs) **102**, user equipments (UEs) **104**, one or more core networks, such as an Evolved Packet Core (EPC) **160** and 5G Core (5GC) network **190**, which interoperate to provide wireless communications services.

(16) Base stations **102** may provide an access point to the EPC **160** and/or 5GC **190** for a user equipment **104**, and may perform one or more of the following functions: transfer of user data, radio channel ciphering and deciphering, integrity protection, header compression, mobility control functions (e.g., handover, dual connectivity), inter-cell interference coordination, connection setup and release, load balancing, distribution for non-access stratum (NAS) messages, NAS node selection, synchronization, radio access network (RAN) sharing, multimedia broadcast multi cast service (MBMS), subscriber and equipment trace, RAN information management (RIM), paging, positioning, delivery of warning messages, among other functions. Base stations may include and/or be referred to as a gNB, NodeB, eNB, ng-eNB (e.g., an eNB that has been enhanced to provide connection to both EPC **160** and 5GC **190**), an access point, a base transceiver station, a radio base station, a radio transceiver, or a transceiver function, or a transmission reception point in various contexts.

(17) Base stations **102** wirelessly communicate with UEs **104** via communications links **120**. Each of base stations **102** may provide communication coverage for a respective geographic coverage area **110**, which may overlap in some cases. For example, small cell **102′** (e.g., a low-power base station) may have a coverage area **110′** that overlaps the coverage area **110** of one or more macrocells (e.g., high-power base stations).

(18) The communication links **120** between base stations **102** and UEs **104** may include uplink (UL) (also referred to as reverse link) transmissions from a user equipment **104** to a base station **102** and/or downlink (DL) (also referred to as forward link) transmissions from a base station **102** to a user equipment **104**. The communication links **120** may use multiple-input and multiple-output (MIMO) antenna technology, including spatial multiplexing, beamforming, and/or transmit diversity in various aspects.

(19) Examples of UEs **104** include a cellular phone, a smart phone, a session initiation protocol (SIP) phone, a laptop, a personal digital assistant (PDA), a satellite radio, a global positioning system, a multimedia device, a video device, a digital audio player, a camera, a game console, a tablet, a smart device, a wearable device, a vehicle, an electric meter, a gas pump, a large or small kitchen appliance, a healthcare device, an implant, a sensor/actuator, a display, or other similar devices. Some of UEs **104** may be internet of things (IoT) devices (e.g., parking meter, gas pump, toaster, vehicles, heart monitor, or other IoT devices), always on (AON) devices, or edge processing devices. UEs **104** may also be referred to more generally as a station, a mobile station, a subscriber station, a mobile unit, a subscriber unit, a wireless unit, a remote unit, a mobile device, a wireless device, a wireless communications device, a remote device, a mobile subscriber station, an access terminal, a mobile terminal, a wireless terminal, a remote terminal, a handset, a user agent, a mobile client, or a client.

(20) Wireless communication network **100** includes uplink transmission configuration component **199**, which may be configured to configure machine learning models for the UE to use in precoding uplink transmissions to the base station. Wireless network **100** further includes uplink

transmission precoding component **198**, which may be used configured to precode uplink transmissions using a precoding identified by machine learning models.

(21) FIG. **2** depicts aspects of an example base station (BS) **102** and a user equipment (UE) **104**.

(22) Generally, base station **102** includes various processors (e.g., **220**, **230**, **238**, and **240**), antennas **234***a-t* (collectively **234**), transceivers **232***a-t* (collectively **232**), which include modulators and demodulators, and other aspects, which enable wireless transmission of data (e.g., source data **212**) and wireless reception of data (e.g., data sink **239**). For example, base station **102** may send and receive data between itself and user equipment **104**.

(23) Base station **102** includes controller/processor **240**, which may be configured to implement various functions related to wireless communications. In the depicted example, controller/processor **240** includes uplink transmission configuration component **241**, which may be representative of uplink transmission configuration component **199** of FIG. **1**. Notably, while depicted as an aspect of controller/processor **240**, uplink transmission configuration component **241** may be implemented additionally or alternatively in various other aspects of base station **102** in other implementations.

(24) Generally, user equipment **104** includes various processors (e.g., **258**, **264**, **266**, and **280**), antennas **252***a-r* (collectively **252**), transceivers **254***a-r* (collectively **254**), which include modulators and demodulators, and other aspects, which enable wireless transmission of data (e.g., source data **262**) and wireless reception of data (e.g., data sink **260**).

(25) User equipment **102** includes controller/processor **280**, which may be configured to implement various functions related to wireless communications. In the depicted example, controller/processor **280** includes uplink transmission precoding component **281**, which may be representative of uplink transmission precoding component **198** of FIG. **1**. Notably, while depicted as an aspect of controller/processor **280**, uplink transmission precoding component **281** may be implemented additionally or alternatively in various other aspects of user equipment **104** in other implementations.

(26) FIGS. **3**A-**3**D depict aspects of data structures for a wireless communication network, such as wireless communication network **100** of FIG. **1**. In particular, FIG. **3**A is a diagram **300** illustrating an example of a first subframe within a 5G (e.g., 5G NR) frame structure, FIG. **3**B is a diagram **330** illustrating an example of DL channels within a 5G subframe, FIG. **3**C is a diagram **350** illustrating an example of a second subframe within a 5G frame structure, and FIG. **3**D is a diagram **380** illustrating an example of UL channels within a 5G subframe.

(27) Further discussions regarding FIG. **1**, FIG. **2**, and FIGS. **3**A-**3**D are provided later in this disclosure.

Aspects Related to Sounding and Transmission Precoding Determination for Uplink Transmissions Using Machine Learning Models

(28) Precoding schemes in multiple user (MU) multiple-input-multiple-output (MIMO) (MU-MIMO) transmission schemes generally allow for transmissions to be beamformed so that a precoded signal transmitted by a transmitting wireless device is received at a receiving wireless device at a higher received power. Generally, the precoding used for transmissions may be determined based on various channel quality measurements. The precoding generally aids in beamforming a transmission such that strength of the transmission, when received by a receiving device, is maximized.

(29) In one example, a precoding can be determined based on channel state reciprocity assumed between an uplink and a downlink channel. Channel state reciprocity generally refers to similarities in channel conditions between uplink and downlink channels. Generally, a network entity may perform channel state measurements on uplink channels (e.g., based on demodulation reference signals (DMRSs) or sounding reference signals (SRSs) to determine a precoder to use for a downlink channel. In using CSI measurements for uplink channels and assumed channel reciprocity for the downlink to determine a precoder to use for a downlink channel, the network entity can assume that the uplink channels and the downlink channels are symmetrical such that

conditions measured on the uplink accurately reflect conditions on the downlink. These assumptions may be made, for example, for convenience, reducing latency entailed by waiting for downlink-specific measurements prior to determining a precoding to use for downlink transmissions, and the like. However, in some cases, the reciprocity assumed between the uplink and the downlink may not hold.

(30) For example, in frequency-domain duplexed (FDD) systems where uplink transmissions are performed using different frequency bands from downlink transmissions, channel conditions on a frequency band used for uplink transmissions may be different than channel conditions on a frequency band for downlink transmissions. Thus, when uplink channel and downlink channel conditions are significantly different (e.g., due to interference on one, but not both, channels), the precoding selected for the downlink channel may be inaccurate. In one example, if the uplink channel has worse channel quality characteristics than the downlink channel and a downlink precoding is selected based on uplink channel measurements and assumed channel reciprocity, radio resources may not be fully used on the downlink (e.g., due to the allocation of additional resources for transmission bundling or other redundancies). In another example, if the uplink channel has better channel quality characteristics than the downlink channel and the precoding is selected based on uplink channel measurements and assumed channel reciprocity, downlink transmissions may fail because the precoding for these downlink transmissions may not be appropriate for the characteristics of the downlink channel, and additional resources may be used for retransmitting failed transmissions on the downlink channel.

(31) Similarly, when uplink channel and downlink channel conditions are significantly different, the precoding selected for the uplink channel may also be inaccurate. In one example, if the uplink channel has worse channel quality characteristics than the downlink channel, and an uplink precoding is selected based on downlink channel measurements and assumed channel reciprocity, uplink transmissions may fail, and additional radio resources and battery power may be used for retransmitting failed transmissions on the uplink channel. If the uplink channel has better channel quality characteristics than the downlink channel, and the uplink precoding is selected based on downlink channel measurements and assumed channel reciprocity, radio resources may not be fully used on the uplink. Thus, in order to transmit a given number of information bits, a larger number of redundancy bits may be transmitted, which may also waste battery power, reduce overall bandwidth, and reduce availability of radio resources for other transmissions.

(32) In another example, precoding matrix indicator-based channel state feedback can be used in MU-MIMO systems to determine precodings for a wireless transmission. In such a case, a network entity can transmit non-precoded CSI-RSs to a UE for measurement. The UE may determine, based on measurements of the non-precoded CSI-RSs, a precoding to use for an uplink transmission to the network entity. Channel state feedback, such as a precoding matrix indicator (PMI), may be precoded using the determined precoding and transmitted to the network entity, and the precoded channel state feedback and non-precoded CSI-RSs may be used to determine a modulation and coding scheme (MCS) for subsequent downlink transmissions. However, multiple rounds of the network entity transmitting non-precoded CSI-RSs to a UE and the UE providing precoded channel state feedback may be needed in order to determine a proper MCS. Thus, a time overhead may be imposed on the determination of a precoding for use in a transmission. Further, channel state feedback may become outdated over time, and thus, the determined MCS may be based on outdated feedback that is no longer applicable for the channel. While this may not be a significant problem in situations where channel conditions remain relatively consistent, this may decrease the applicability of the determined MCS in situations where channel conditions rapidly change (e.g. when the UE is mobility, or when radio conditions are changing around the UE due to weather-induced radio propagation changes, interference introduced from other devices, or the like).

(33) In still another example, explicit CSI feedback may be used to determine a precoding for a wireless transmission. In this case, a network entity can transmit CSI-RS s for a UE to measure,

and the UE can measure CSI based on the CSI-RSs and report the measured CSI to the network entity for use in determining a precoding. A large amount of data may be transmitted in order to provide a network entity a good representation of the CSI, which may increase the overhead entailed in sounding and precoding determination.

(34) To improve the process of determining precodings to use for downlink transmissions between a transmitting device and a receiving device (e.g., between a network entity and a UE), various machine learning-based techniques can be used.

(35) For example, in a deep neural network (DNN)-based downlink precoding scheme, DNNs may be trained based on information for a single UE for common channel state feedback. These DNNs may include, for example, a CSI-RS DNN deployed at a network entity and trained to generate a CSI-RS pattern for CSI-RSs to be transmitted to a UE for measurement, a channel state feedback DNN deployed at the UE to process the received CSI-RS pattern and to generate a feedback report, and a single user (SU) precoding DNN deployed at the network entity and trained to determine a single user precoding based on the received feedback. Each of the first, second, and third DNNs may be trained based on a loss function that maximizes a transmission and/or reception metric, such as a signal-plus-interference-to-noise ratio (SINR).

(36) In some cases, each UE with which the network entity communicates can use an instance of the trained CSI-RS DNN and channel state feedback DNN, and a multi-user precoding DNN can be trained at the network side for use in determining a precoding to use for downlink transmissions. The multi-user precoding DNN may be trained, for example, based on a loss function that maximizes sum rate, or the number of bits transmitted per unit of energy consumed by the downlink transmission.

(37) The DNN-based downlink precoding schemes may allow for reductions in payload size relative to explicit channel state feedback. For example, for a given number of information bits to be included in a downlink transmission, the total number of bits transmitted may be lower using DNN-based downlink precoding schemes as opposed to downlink transmissions using precodings selected using explicit channel state feedback. Additionally, UE complexity may be reduced due to the replacement of channel estimation with a neural network-based solution, and the network entity may replace the complexity of multi-user precoding calculations with less computationally expensive determinations based on the output of the multi-user precoding deep neural network. However, while DNN-based downlink precoding schemes may provide for improvements in determining precodings used for downlink transmissions, these DNN-based downlink precoding schemes may not provide these improvements for uplink transmissions.

(38) Aspects of the present disclosure provide techniques for machine learning-based sounding and precoding for uplink transmissions. The techniques may be used, for example, to increase the accuracy or applicability of precodings used for uplink transmissions and to reduce processing overhead in determining an appropriate precoding for an uplink transmission, and may thus result in more efficient use of wireless communications resources, improved reliability for communications between the network entity and the UE, power savings due to a reduction in the amount of data transmitted and/or received in determining a precoding for uplink transmissions and in performing re-transmission of failed transmissions, and the like. Further, aspects of the present disclosure account for the variability of UE antenna configurations. A unified transmission precoding matrix indicator (TPMI) codebook for a universe of unique UEs need not be defined in order to allow for UE-specific sounding and uplink precoding techniques.

(39) FIG. **4** illustrates a call flow diagram of messages that may be exchanged between a UE **402** and a base station **406** to precode an uplink transmission based on SRSs and precoding matrices generated or otherwise identified by one or more machine learning models. An SRS DNN **403** and a precoding matrix interpretation DNN **404** may be deployed on UE **402**, and a precoding matrix DNN **407** may be deployed on base station **406**.

(40) As illustrated, base station **406** can transmit configuration information **410** to UE **402**.

Configuration information **410** may include, for example, configuration information for the SRS DNN **403** and the precoding matrix DNN **404**. Generally, the configuration information may include values of weight and/or bias parameters for these DNNs, which may be defined by the base station **406** or statically defined (e.g., in a wireless communication standard).

(41) At block **412**, the UE **402** configures the SRS DNN **403** and the precoding matrix DNN **404** based on the received configuration information.

(42) Generally, to configure the SRS DNN **403** and the precoding matrix DNN **404**, the UE can set values for one or more parameters of these DNNs to the values of the associated parameters in the received configuration information. In some aspects, where the received configuration information includes indices of sets of parameters in a lookup table, the UE can configure the SRS DNN **403** and/or precoding matrix DNN **404** by specifying the index in the parameters of the SRS DNN **403** and/or precoding matrix DNN **404** or by retrieving the weight and/or bias parameters from the lookup table and configuring the SRS DNN **403** and/or precoding matrix DNN **404** with the retrieved weight and/or bias parameters.

(43) It should be understood that, while the configuration information is illustrated as being communicated from base station **406** to UE **402**, the SRS DNN and the precoding matrix DNN may be configured a priori at the UE **402** and need not be configured by the base station **406**.

(44) At block **414**, the UE **402** generates an SRS using the SRS DNN. The inputs into the SRS DNN **403** may be indicated by the network entity (e.g., in configuration information received from the network entity) or may be defined a priori (e.g., in a wireless communication standard). The inputs may include, for example, an indication of a base sequence, subframe configuration information, bandwidth information, cyclic shift information, and/or other data that can be used as a basis for generating the SRS. The weights and biases of the SRS DNN **403** may also be indicated by the network entity in the configuration information received from the network entity or defined a priori, as discussed above. The output of the SRS DNN **403** may be a generated SRS, and the generated SRS may be transmitted to the network entity for use in identifying a precoding matrix for the UE **402** to use for uplink transmissions on a shared channel.

(45) At block **418**, the base station **406** identifies a precoding matrix based on the SRS. Generally, to identify the precoding matrix, the base station **406** can use the SRS **416** as input into the precoding matrix DNN **407**. The precoding matrix DNN **407** generally uses at least the received SRS as input to identify a precoding matrix for the UE **402** to use for precoding a subsequent uplink transmission to the network entity. The weights and biases of the precoding matrix DNN **407** may be, for example, defined a priori (e.g., in a wireless communication standard) or provided to the network entity from a core network management entity. The precoding matrix DNN **407** can generate a transmission precoding matrix indicator (TPMI) or other information identifying the properties of a precoding matrix that the UE **402** is to use for precoding the uplink transmission, and the output of the precoding matrix DNN **407** may be transmitted to the UE **402** for use in precoding the uplink transmission.

(46) At block **422**, the UE **402** precodes uplink transmissions based on the identified precoding matrix. To precode an uplink transmission, the UE **402** can use the received precoding matrix identification **420** as input into a precoding matrix interpretation DNN **404**, which may be trained to identify a precoding to apply to an uplink transmission based on a received precoding matrix identification. The UE **402** can then precode the uplink transmission and transmit the precoded uplink transmission **424** to the base station **406**. The precoding matrix interpretation DNN **404** generally uses at least the received information identifying the precoding matrix to apply a precoding to an uplink transmission. In some aspects, the precoding matrix interpretation Precoding matrix interpretation DNN **404** may be trained to map the received information identifying the precoding matrix (which may include, for example a TPMI generated by the precoding matrix DNN **407**) to a precoding to apply to the uplink transmission. The weights and biases of the precoding matrix interpretation DNN **404** may be configured by a network entity or

defined a priori (e.g., in a wireless communication standard).
(47) FIG. **5** illustrates example operations **500** for wireless communications by a user equipment (UE). Operations **500** may be performed, for example, by uplink transmission precoding component **198** or uplink transmission precoding component **281** illustrated in FIG. **1** or **2**.
(48) As illustrated, operations **500** begin at block **510**, where the UE generates a sounding reference signal (SRS) using an SRS deep neural network (DNN) (e.g., SRS DNN **403** illustrated in FIG. **4**). In some aspects, the weights and biases for the SRS DNN may include a set of weights and biases defined apriori (e.g., in a wireless communication standard). In some aspects, the weights and biases for the SRS DNN may be configured by the network entity. In such a case, the UE may receive, from the network entity, information identifying these parameters. The information may include explicitly communicated values for one or more bias and/or weight parameters of the SRS DNN. In some cases, the parameters may be, for example, bias and/or weight parameters selected from sets of parameters in a lookup table, where each set of parameters is associated with a unique index in the lookup table. The configuration information communicated to the UE may include an index in the lookup table, and using the index, the UE can identify the set of bias and/or weight parameters to use for the SRS DNN.
(49) At block **520**, the UE transmits, to a network entity, the generated SRS in one or more resource elements (REs).
(50) In some aspects, the one or more REs, may be statically defined REs. For example, the statically defined REs may be defined in a wireless communication standard or otherwise defined a priori. In some aspects, the UE can receive configuration information from the network entity identifying the one or more REs.
(51) In some aspects, the UE can apply a code division multiplexing (CDM) cover code to the generated SRS. After applying to the CDM cover code to the generated SRS, the generated SRS may be transmitted to the network entity.
(52) At block **530**, the UE receives, from the network entity, information identifying a precoding matrix to use for uplink transmissions on a shared channel. For example, the information identifying the precoding matrix may include a precoding matrix identifier (PMI), a transmission precoding matrix identifier (TPMI), or the like.
(53) At block **540**, the UE precodes uplink transmissions based on the identified precoding matrix.
(54) At block **550**, the UE transmits, to the network entity, the precoded uplink transmissions on the shared channel.
(55) In some aspects, the UE can receive, from the network entity, a transmitted precoding matrix indicator (TPMI), based on the generated SRS, for use in precoding the uplink transmissions. For example, the UE can precode the uplink transmissions by applying a precoding to the uplink transmissions based, at least in part, on an output from one or more precoding matrix interpretation DNNs (e.g., precoding matrix interpretation DNN **404** illustrated in FIG. **4**). In such examples, the TPMI received from the network entity may be used as an input into the precoding DNNs. In some aspects, the parameters of the one or more precoding DNNs may be defined in a wireless communication standard or otherwise statically defined a priori. In some aspects, the parameters of the one or more precoding DNNs may be configured via configuration information received from the network entity.
(56) In some aspects, the UE can precode the uplink transmissions by applying a precoding based on an association between the received TPMI and a precoding to apply to the uplink transmissions. In some aspects, the association between the received TPMI and a precoding to apply to the uplink transmissions may be statically defined (e.g., in a wireless communication standard) or otherwise defined a priori. The association between the received TPMI and the precoding to apply to the uplink transmissions may be identified in configuration information received from the network entity. The mapping may be determined by one or more precoding matrix interpretation DNNs, which may be configured to identify a precoding to apply to the uplink transmission based on a

probability distribution over a universe of precodings. For example, the one or more precoding matrix interpretation DNNs may use a softmax layer over a universe of precodings to generate a probability distribution over the universe of precodings. The precoding associated with a highest probability in the probability distribution may be selected for use in precoding the uplink transmissions.

(57) FIG. **6** illustrates example operations **600** that may be performed by a network entity (e.g., a gNodeB) to receive precoded uplink transmissions from a UE based on precoding matrix information determined based on one or more machine learning models, according to aspects of the present disclosure. Operations **600** may be performed, for example, by uplink transmission configuration component **199** or uplink transmission configuration component **241** illustrated in FIG. **1** or **2** and may be complementary to operations **500** discussed above.

(58) Operations **600**, as illustrated, begin at block **610**, where the network entity receives, from a UE, a sounding reference signal. As discussed, the sounding reference signal may be generated by an SRS DNN, the parameters of which may be indicated to the UE by the network entity or may be statically defined (e.g., in a wireless communication standard).

(59) At block **620**, the network entity can identify a precoding matrix for the UE to use in uplink transmissions based on a precoding matrix DNN (e.g., precoding matrix DNN **407** illustrated in FIG. **4**) and the received SRS. Generally, the precoding matrix DNN may be trained to learn relationships between a received SRS (e.g., the contents of the received SRS, measurements performed based on receiving the SRS, etc.) and a transmission precoding matrix that may result in receipt of uplink transmissions from the UE with a highest signal strength. In some aspects, the information identifying the precoding matrix may include one or more parameters that a precoding DNN can use to identify the precoding to be used for uplink transmissions to the network entity and need not, for example, explicitly identify the precoding.

(60) At block **630**, the network entity transmits, to the UE, information identifying the precoding matrix.

(61) At block **640**, the network entity receives uplink transmissions precoded using the identified precoding matrix.

Example Wireless Communication Devices

(62) FIG. **7** depicts an example communications device **700** that includes various components operable, configured, or adapted to perform operations for the techniques disclosed herein, such as the operations depicted and described with respect to FIG. **5**. In some examples, communication device **700** may be a user equipment **104** as described, for example with respect to FIGS. **1** and **2**.

(63) Communications device **700** includes a processing system **702** coupled to a transceiver **708** (e.g., a transmitter and/or a receiver). Transceiver **708** is configured to transmit (or send) and receive signals for the communications device **700** via an antenna **710**, such as the various signals as described herein. Processing system **702** may be configured to perform processing functions for communications device **700**, including processing signals received and/or to be transmitted by communications device **700**.

(64) Processing system **702** includes one or more processors **720** coupled to a computer-readable medium/memory **720** via a bus **706**. In certain aspects, computer-readable medium/memory **720** is configured to store instructions (e.g., computer-executable code) that when executed by the one or more processors **720**, cause the one or more processors **720** to perform the operations illustrated in FIG. **5**, or other operations for performing the various techniques discussed herein to precode uplink transmissions using a precoding identified by machine learning models.

(65) In the depicted example, computer-readable medium/memory **730** stores code **731** for generating a sounding reference signal (SRS), code **732** for transmitting the generated SRS, code **733** for receiving information identifying a precoding matrix to use for uplink transmissions, code **734** for precoding uplink transmissions based on the identified precoding matrix, and code **735** for transmitting the precoded uplink transmissions.

(66) In the depicted example, the one or more processors **720** include circuitry configured to implement the code stored in the computer-readable medium/memory **720**, including circuitry **721** for generating a sounding reference signal (SRS), circuitry **722** for transmitting the generated SRS, circuitry **723** for receiving information identifying a precoding matrix to use for uplink transmissions, circuitry **724** for precoding uplink transmissions based on the identified precoding matrix, circuitry **725** for transmitting the precoded uplink transmissions.
(67) Various components of communications device **700** may provide means for performing the methods described herein, including with respect to FIG. **5**.
(68) In some examples, means for transmitting or sending (or means for outputting for transmission) may include the transceivers **254** and/or antenna(s) **252** of the user equipment **104** illustrated in FIG. **2** and/or transceiver **708** and antenna **710** of the communication device **700** in FIG. **7**.
(69) In some examples, means for receiving (or means for obtaining) may include the transceivers **254** and/or antenna(s) **252** of the user equipment **104** illustrated in FIG. **2** and/or transceiver **708** and antenna **710** of the communication device **700** in FIG. **7**.
(70) In some examples, means for generating a sounding reference signal and means for precoding uplink transmissions may include various processing system components, such as: the one or more processors **720** in FIG. **7**, or aspects of the user equipment **104** depicted in FIG. **2**, including receive processor **258**, transmit processor **264**, TX MIMO processor **266**, and/or controller/processor **280** (including uplink transmission precoding component **281**).
(71) Notably, FIG. **7** is just use example, and many other examples and configurations of communication device **700** are possible.
(72) FIG. **8** depicts an example communications device **800** that includes various components operable, configured, or adapted to perform operations for the techniques disclosed herein, such as the operations depicted and described with respect to FIG. **6**. In some examples, communication device **800** may be a base station **102** as described, for example with respect to FIGS. **1** and **2**.
(73) Communications device **800** includes a processing system **802** coupled to a transceiver **808** (e.g., a transmitter and/or a receiver). Transceiver **808** is configured to transmit (or send) and receive signals for the communications device **800** via an antenna **810**, such as the various signals as described herein. Processing system **802** may be configured to perform processing functions for communications device **800**, including processing signals received and/or to be transmitted by communications device **800**.
(74) Processing system **802** includes one or more processors **820** coupled to a computer-readable medium/memory **820** via a bus **806**. In certain aspects, computer-readable medium/memory **820** is configured to store instructions (e.g., computer-executable code) that when executed by the one or more processors **820**, cause the one or more processors **820** to perform the operations illustrated in FIG. **6**, or other operations for performing the various techniques discussed herein to configure machine learning models for the UE to use in precoding uplink transmissions to the base station.
(75) In the depicted example, computer-readable medium/memory **830** stores code **831** for receiving a sounding reference signal, code **832** for identifying a precoding matrix, code **833** for transmitting the information identifying the precoding matrix, and code **834** for receiving uplink transmissions precoded using the identified precoding matrix.
(76) In the depicted example, the one or more processors **820** include circuitry configured to implement the code stored in the computer-readable medium/memory **820**, including circuitry **821** for receiving a sounding reference signal, circuitry **822** for identifying a precoding matrix, circuitry **823** for transmitting the information identifying the precoding matrix, and circuitry **824** for receiving uplink transmissions precoded using the identified precoding matrix.
(77) Various components of communications device **800** may provide means for performing the methods described herein, including with respect to FIG. **6**.
(78) In some examples, means for transmitting or sending (or means for outputting for

transmission) may include the transceivers **232** and/or antenna(s) **234** of the base station **102** illustrated in FIG. **2** and/or transceiver **808** and antenna **810** of the communication device **800** in FIG. **8**.

(79) In some examples, means for receiving (or means for obtaining) may include the transceivers **232** and/or antenna(s) **234** of the base station illustrated in FIG. **2** and/or transceiver **808** and antenna **810** of the communication device **800** in FIG. **8**.

(80) In some examples, means for identifying a precoding matrix may include various processing system components, such as: the one or more processors **820** in FIG. **8**, or aspects of the base station **102** depicted in FIG. **2**, including receive processor **238**, transmit processor **220**, TX MIMO processor **230**, and/or controller/processor **240** (including uplink transmission configuration component **241**).

(81) Notably, FIG. **8** is just use example, and many other examples and configurations of communication device **800** are possible.

Example Clauses

(82) Implementation examples are described in the following numbered clauses:

(83) Clause 1: A method for wireless communications by a user equipment (UE), comprising: generating a sounding reference signal (SRS) using an SRS deep neural network (DNNs); transmitting, to a network entity, the generated SRS in one or more resource elements (REs); receiving, from the network entity, information identifying a precoding matrix to use for uplink transmissions on a shared channel; precoding uplink transmissions based on the identified precoding matrix; and transmitting, to the network entity, the precoded uplink transmissions on the shared channel.

(84) Clause 2: The method of Clause 1, wherein parameters of the SRS DNN comprise a set of weights and biases defined in a wireless communication standard.

(85) Clause 3: The method of any one of Clauses 1 or 2, further comprising receiving, from the network entity, configuration information identifying parameters of the SRS DNN.

(86) Clause 4: The method of Clause 3, wherein: the configuration information identifying parameters of the SRS DNNs comprises an index in a lookup table, and the index in the lookup table is mapped to a set of bias and weight parameters for the SRS DNN.

(87) Clause 5: The method of Clause 3, wherein the configuration information includes values for one or more bias and weight parameters of the SRS DNN.

(88) Clause 6: The method of any one of Clauses 1 through 5, wherein the generated SRS is transmitted on one or more statically defined resource elements.

(89) Clause 7: The method of any one of Clauses 1 through 5, further comprising receiving, from the network entity, configuration information identifying the one or more REs.

(90) Clause 8: The method of any one of Clauses 1 through 7, further comprising applying a code division multiplexing (CDM) cover code to the generated SRS prior to transmitting the generated SRS to the network entity.

(91) Clause 9: The method of any one of Clauses 1 through 8, further comprising: receiving, from the network entity, a transmitted precoding matrix indicator (TPMI), based on the generated SRS, for use in precoding the uplink transmissions, wherein precoding the uplink transmissions comprises: applying a precoding to the uplink transmissions based, at least in part, on one or more precoding DNNs, given the TPMI as input.

(92) Clause 10: The method of Clause 9, wherein parameters of the one or more precoding DNNs comprise a statically defined set of weights and biases.

(93) Clause 11: The method of Clause 9, further comprising: receiving, from the network entity, configuration information identifying parameters of the one or more precoding DNNs.

(94) Clause 12: The method of any one of Clauses 1 through 11, further comprising: receiving, from the network entity, a transmitted precoding matrix indicator (TPMI) for use in precoding uplink transmissions, wherein precoding the uplink transmissions comprises applying a precoding

based on an association between the received TPMI and a precoding to apply to the uplink transmissions.
(95) Clause 13: The method of Clause 12, further comprising receiving, from the network entity, configuration information identifying the association between the received TPMI and the precoding to apply to the uplink transmissions.
(96) Clause 14: The method of Clause 13, wherein the configuration information identifying the association between the received TPMI and the precoding information comprises a probabilistic model configured to identify a precoding to apply to the uplink transmissions based on a probability distribution over a universe of precodings, given the received TPMI as input.
(97) Clause 15: The method of Clause 12, wherein the association between the received TPMI and the precoding to apply to the uplink transmissions comprises a statically defined association.
(98) Clause 16: A method for wireless communications by a network entity, comprising: receiving, from a user equipment (UE), a sounding reference signal (SRS) generated by an SRS deep neural network (DNN); identifying a precoding matrix based on a precoding matrix deep neural network and the received SRS; transmitting, to the UE, the information identifying the precoding matrix; and receiving, from the UE, uplink transmissions precoded using the identified precoding matrix.
(99) Clause 17: The method of Clause 16, further comprising transmitting, to the UE, configuration information identifying parameters of the SRS DNN.
(100) Clause 18: The method of Clause 17, wherein: the configuration information identifying parameters of the SRS DNN comprises an index in a lookup table, and the index in the lookup table is mapped to a set of bias and weight parameters for the SRS DNN.
(101) Clause 19: The method of Clause 17, wherein the configuration information includes values for one or more bias and weight parameters of the SRS DNN.
(102) Clause 20: The method of any one of Clauses 16 through 19, further comprising transmitting, to the UE, configuration information identifying one or more REs on which the SRS is to be received.
(103) Clause 21: The method of any one of Clauses 16 through 20, further comprising transmitting, to the UE, a transmitted precoding matrix indicator (TPMI), based on the generated SRS, for use in precoding the uplink transmissions via one or more precoding DNNs.
(104) Clause 22: The method of Clause 21, further comprising transmitting, to the UE, configuration information identifying parameters of the one or more precoding DNNs.
(105) Clause 23: The method of Clause 21, further comprising transmitting, to the UE, configuration information identifying an association between a TPMI and a precoding to be applied to the uplink transmissions.
(106) Clause 24: The method of Clause 23, wherein the configuration information identifying the association between the received TPMI and the precoding information comprises a probabilistic model configured to identify a precoding to apply to the uplink transmissions based on a probability distribution over a universe of precodings, given the received TPMI as input.
(107) Clause 25: A processing system, comprising: a memory comprising computer-executable instructions; and one or more processors configured to execute the computer-executable instructions and cause the processing system to perform a method in accordance with any one of Clauses 1-24.
(108) Clause 26: A processing system, comprising: means for performing a method in accordance with any one of Clauses 1-24.
(109) Clause 27: A non-transitory computer-readable medium comprising computer-executable instructions that, when executed by one or more processors of a processing system, cause the processing system to perform a method in accordance with any one of Clauses 1-24.
(110) Clause 28: A computer program product embodied on a computer-readable storage medium comprising code for performing a method in accordance with any one of Clauses 1-24.
Additional Wireless Communication Network Considerations

(111) The techniques and methods described herein may be used for various wireless communications networks (or wireless wide area network (WWAN)) and radio access technologies (RATs). While aspects may be described herein using terminology commonly associated with 3G, 4G, and/or 5G (e.g., 5G new radio (NR)) wireless technologies, aspects of the present disclosure may likewise be applicable to other communication systems and standards not explicitly mentioned herein.
(112) 5G wireless communication networks may support various advanced wireless communication services, such as enhanced mobile broadband (eMBB), millimeter wave (mmWave), machine type communications (MTC), and/or mission critical targeting ultra-reliable, low-latency communications (URLLC). These services, and others, may include latency and reliability requirements.
(113) Returning to FIG. **1**, various aspects of the present disclosure may be performed within the example wireless communication network **100**.
(114) In 3GPP, the term "cell" can refer to a coverage area of a NodeB and/or a narrowband subsystem serving this coverage area, depending on the context in which the term is used. In NR systems, the term "cell" and BS, next generation NodeB (gNB or gNodeB), access point (AP), distributed unit (DU), carrier, or transmission reception point may be used interchangeably. A BS may provide communication coverage for a macro cell, a pico cell, a femto cell, and/or other types of cells.
(115) A macro cell may generally cover a relatively large geographic area (e.g., several kilometers in radius) and may allow unrestricted access by UEs with service subscription. A pico cell may cover a relatively small geographic area (e.g., a sports stadium) and may allow unrestricted access by UEs with service subscription. A femto cell may cover a relatively small geographic area (e.g., a home) and may allow restricted access by UEs having an association with the femto cell (e.g., UEs in a Closed Subscriber Group (CSG) and UEs for users in the home). A BS for a macro cell may be referred to as a macro BS. A BS for a pico cell may be referred to as a pico BS. A BS for a femto cell may be referred to as a femto BS, home BS, or a home NodeB.
(116) Base stations **102** configured for 4G LTE (collectively referred to as Evolved Universal Mobile Telecommunications System (UMTS) Terrestrial Radio Access Network (E-UTRAN)) may interface with the EPC **160** through first backhaul links **132** (e.g., an S1 interface). Base stations **102** configured for 5G (e.g., 5G NR or Next Generation RAN (NG-RAN)) may interface with 5GC **190** through second backhaul links **184**. Base stations **102** may communicate directly or indirectly (e.g., through the EPC **160** or 5GC **190**) with each other over third backhaul links **134** (e.g., X2 interface). Third backhaul links **134** may generally be wired or wireless.
(117) Small cell **102′** may operate in a licensed and/or an unlicensed frequency spectrum. When operating in an unlicensed frequency spectrum, the small cell **102′** may employ NR and use the same 5 GHz unlicensed frequency spectrum as used by the Wi-Fi AP **150**. Small cell **102′**, employing NR in an unlicensed frequency spectrum, may boost coverage to and/or increase capacity of the access network.
(118) Some base stations, such as gNB **180** may operate in a traditional sub-6 GHz spectrum, in millimeter wave (mmWave) frequencies, and/or near mmWave frequencies in communication with the UE **104**. When the gNB **180** operates in mmWave or near mmWave frequencies, the gNB **180** may be referred to as an mmWave base station.
(119) The communication links **120** between base stations **102** and, for example, UEs **104**, may be through one or more carriers. For example, base stations **102** and UEs **104** may use spectrum up to Y MHz (e.g., 5, 10, 15, 20, 100, 400, and other MHz) bandwidth per carrier allocated in a carrier aggregation of up to a total of Yx MHz (x component carriers) used for transmission in each direction. The carriers may or may not be adjacent to each other. Allocation of carriers may be asymmetric with respect to DL and UL (e.g., more or fewer carriers may be allocated for DL than for UL). The component carriers may include a primary component carrier and one or more

secondary component carriers. A primary component carrier may be referred to as a primary cell (PCell) and a secondary component carrier may be referred to as a secondary cell (SCell).
(120) Wireless communications system **100** further includes a Wi-Fi access point (AP) **150** in communication with Wi-Fi stations (STAs) **152** via communication links **154** in, for example, a 2.4 GHz and/or 5 GHz unlicensed frequency spectrum. When communicating in an unlicensed frequency spectrum, the STAs **152**/AP **150** may perform a clear channel assessment (CCA) prior to communicating in order to determine whether the channel is available.
(121) Certain UEs **104** may communicate with each other using device-to-device (D2D) communication link **158**. The D2D communication link **158** may use the DL/UL WWAN spectrum. The D2D communication link **158** may use one or more sidelink channels, such as a physical sidelink broadcast channel (PSBCH), a physical sidelink discovery channel (PSDCH), a physical sidelink shared channel (PSSCH), and a physical sidelink control channel (PSCCH). D2D communication may be through a variety of wireless D2D communications systems, such as for example, FlashLinQ, WiMedia, Bluetooth, ZigBee, Wi-Fi based on the IEEE 802.11 standard, 4G (e.g., LTE), or 5G (e.g., NR), to name a few options.
(122) EPC **160** may include a Mobility Management Entity (MME) **162**, other MMEs **164**, a Serving Gateway **166**, a Multimedia Broadcast Multicast Service (MBMS) Gateway **168**, a Broadcast Multicast Service Center (BM-SC) **170**, and a Packet Data Network (PDN) Gateway **172**. MME **162** may be in communication with a Home Subscriber Server (HSS) **174**. MME **162** is the control node that processes the signaling between the UEs **104** and the EPC **160**. Generally, MME **162** provides bearer and connection management.
(123) Generally, user Internet protocol (IP) packets are transferred through Serving Gateway **166**, which itself is connected to PDN Gateway **172**. PDN Gateway **172** provides UE IP address allocation as well as other functions. PDN Gateway **172** and the BM-SC **170** are connected to the IP Services **176**, which may include, for example, the Internet, an intranet, an IP Multimedia Subsystem (IMS), a PS Streaming Service, and/or other IP services.
(124) BM-SC **170** may provide functions for MBMS user service provisioning and delivery. BM-SC **170** may serve as an entry point for content provider MBMS transmission, may be used to authorize and initiate MBMS Bearer Services within a public land mobile network (PLMN), and may be used to schedule MBMS transmissions. MBMS Gateway **168** may be used to distribute MBMS traffic to the base stations **102** belonging to a Multicast Broadcast Single Frequency Network (MBSFN) area broadcasting a particular service, and may be responsible for session management (start/stop) and for collecting eMBMS related charging information.
(125) 5GC **190** may include an Access and Mobility Management Function (AMF) **192**, other AMFs **193**, a Session Management Function (SMF) **194**, and a User Plane Function (UPF) **195**. AMF **192** may be in communication with a Unified Data Management (UDM) **196**.
(126) AMF **192** is generally the control node that processes the signaling between UEs **104** and 5GC **190**. Generally, AMF **192** provides QoS flow and session management.
(127) All user Internet protocol (IP) packets are transferred through UPF **195**, which is connected to the IP Services **197**, and which provides UE IP address allocation as well as other functions for 5GC **190**. IP Services **197** may include, for example, the Internet, an intranet, an IP Multimedia Subsystem (IMS), a PS Streaming Service, and/or other IP services.
(128) Returning to FIG. **2**, various example components of BS **102** and UE **104** (e.g., the wireless communication network **100** of FIG. **1**) are depicted, which may be used to implement aspects of the present disclosure.
(129) At BS **102**, a transmit processor **220** may receive data from a data source **212** and control information from a controller/processor **240**. The control information may be for the physical broadcast channel (PBCH), physical control format indicator channel (PCFICH), physical hybrid ARQ indicator channel (PHICH), physical downlink control channel (PDCCH), group common PDCCH (GC PDCCH), and others. The data may be for the physical downlink shared channel

(PDSCH), in some examples.
(130) A medium access control (MAC)-control element (MAC-CE) is a MAC layer communication structure that may be used for control command exchange between wireless nodes. The MAC-CE may be carried in a shared channel such as a physical downlink shared channel (PDSCH), a physical uplink shared channel (PUSCH), or a physical sidelink shared channel (PSSCH).
(131) Processor **220** may process (e.g., encode and symbol map) the data and control information to obtain data symbols and control symbols, respectively. Transmit processor **220** may also generate reference symbols, such as for the primary synchronization signal (PSS), secondary synchronization signal (SSS), PBCH demodulation reference signal (DMRS), and channel state information reference signal (CSI-RS).
(132) Transmit (TX) multiple-input multiple-output (MIMO) processor **230** may perform spatial processing (e.g., precoding) on the data symbols, the control symbols, and/or the reference symbols, if applicable, and may provide output symbol streams to the modulators (MODs) in transceivers **232***a*-**232***t*. Each modulator in transceivers **232***a*-**232***t* may process a respective output symbol stream (e.g., for OFDM) to obtain an output sample stream. Each modulator may further process (e.g., convert to analog, amplify, filter, and upconvert) the output sample stream to obtain a downlink signal. Downlink signals from the modulators in transceivers **232***a*-**232***t* may be transmitted via the antennas **234***a*-**234***t*, respectively.
(133) At UE **104**, antennas **252***a*-**252***r* may receive the downlink signals from the BS **102** and may provide received signals to the demodulators (DEMODs) in transceivers **254***a*-**254***r*, respectively. Each demodulator in transceivers **254***a*-**254***r* may condition (e.g., filter, amplify, downconvert, and digitize) a respective received signal to obtain input samples. Each demodulator may further process the input samples (e.g., for OFDM) to obtain received symbols.
(134) MIMO detector **256** may obtain received symbols from all the demodulators in transceivers **254***a*-**254***r*, perform MIMO detection on the received symbols if applicable, and provide detected symbols. Receive processor **258** may process (e.g., demodulate, deinterleave, and decode) the detected symbols, provide decoded data for the UE **104** to a data sink **260**, and provide decoded control information to a controller/processor **280**.
(135) On the uplink, at UE **104**, transmit processor **264** may receive and process data (e.g., for the physical uplink shared channel (PUSCH)) from a data source **262** and control information (e.g., for the physical uplink control channel (PUCCH) from the controller/processor **280**. Transmit processor **264** may also generate reference symbols for a reference signal (e.g., for the sounding reference signal (SRS)). The symbols from the transmit processor **264** may be precoded by a TX MIMO processor **266** if applicable, further processed by the modulators in transceivers **254***a*-**254***r* (e.g., for SC-FDM), and transmitted to BS **102**.
(136) At BS **102**, the uplink signals from UE **104** may be received by antennas **234***a-t*, processed by the demodulators in transceivers **232***a*-**232***t*, detected by a MIMO detector **236** if applicable, and further processed by a receive processor **238** to obtain decoded data and control information sent by UE **104**. Receive processor **238** may provide the decoded data to a data sink **239** and the decoded control information to the controller/processor **240**.
(137) Memories **242** and **282** may store data and program codes for BS **102** and UE **104**, respectively.
(138) Scheduler **244** may schedule UEs for data transmission on the downlink and/or uplink.
(139) 5G may utilize orthogonal frequency division multiplexing (OFDM) with a cyclic prefix (CP) on the uplink and downlink. 5G may also support half-duplex operation using time division duplexing (TDD). OFDM and single-carrier frequency division multiplexing (SC-FDM) partition the system bandwidth into multiple orthogonal subcarriers, which are also commonly referred to as tones and bins. Each subcarrier may be modulated with data. Modulation symbols may be sent in the frequency domain with OFDM and in the time domain with SC-FDM. The spacing between adjacent subcarriers may be fixed, and the total number of subcarriers may be dependent on the

system bandwidth. The minimum resource allocation, called a resource block (RB), may be 12 consecutive subcarriers in some examples. The system bandwidth may also be partitioned into subbands. For example, a subband may cover multiple RBs. NR may support a base subcarrier spacing (SCS) of 15 KHz and other SCS may be defined with respect to the base SCS (e.g., 30 kHz, 60 kHz, 120 kHz, 240 kHz, and others).
(140) As above, FIGS. **3**A-**3**D depict various example aspects of data structures for a wireless communication network, such as wireless communication network **100** of FIG. **1**.
(141) In various aspects, the 5G frame structure may be frequency division duplex (FDD), in which for a particular set of subcarriers (carrier system bandwidth), subframes within the set of subcarriers are dedicated for either DL or UL. 5G frame structures may also be time division duplex (TDD), in which for a particular set of subcarriers (carrier system bandwidth), subframes within the set of subcarriers are dedicated for both DL and UL. In the examples provided by FIGS. **3**A and **3**C, the 5G frame structure is assumed to be TDD, with subframe 4 being configured with slot format 28 (with mostly DL), where D is DL, U is UL, and X is flexible for use between DL/UL, and subframe 3 being configured with slot format 34 (with mostly UL). While subframes 3, 4 are shown with slot formats 34, 28, respectively, any particular subframe may be configured with any of the various available slot formats 0-61. Slot formats 0, 1 are all DL, UL, respectively. Other slot formats 2-61 include a mix of DL, UL, and flexible symbols. UEs are configured with the slot format (dynamically through DL control information (DCI), or semi-statically/statically through radio resource control (RRC) signaling) through a received slot format indicator (SFI). Note that the description below applies also to a 5G frame structure that is TDD.
(142) Other wireless communication technologies may have a different frame structure and/or different channels. A frame (10 ms) may be divided into 10 equally sized subframes (1 ms). Each subframe may include one or more time slots. Subframes may also include mini-slots, which may include 7, 4, or 2 symbols. In some examples, each slot may include 7 or 14 symbols, depending on the slot configuration.
(143) For example, for slot configuration 0, each slot may include 14 symbols, and for slot configuration 1, each slot may include 7 symbols. The symbols on DL may be cyclic prefix (CP) OFDM (CP-OFDM) symbols. The symbols on UL may be CP-OFDM symbols (for high throughput scenarios) or discrete Fourier transform (DFT) spread OFDM (DFT-s-OFDM) symbols (also referred to as single carrier frequency-division multiple access (SC-FDMA) symbols) (for power limited scenarios; limited to a single stream transmission).
(144) The number of slots within a subframe is based on the slot configuration and the numerology. For slot configuration 0, different numerologies 0 to 5 allow for 1, 2, 4, 8, 16, and 32 slots, respectively, per subframe. For slot configuration 1, different numerologies 0 to 2 allow for 2, 4, and 8 slots, respectively, per subframe. Accordingly, for slot configuration 0 and numerology μ, there are 14 symbols/slot and 2μ slots/subframe. The subcarrier spacing and symbol length/duration are a function of the numerology. The subcarrier spacing may be equal to 2.sup.μ×15 kHz, where μ is the numerology 0 to 5. As such, the numerology μ=0 has a subcarrier spacing of 15 kHz and the numerology μ=5 has a subcarrier spacing of 480 kHz. The symbol length/duration is inversely related to the subcarrier spacing. FIGS. **3**A-**3**D provide an example of slot configuration 0 with 14 symbols per slot and numerology μ=2 with 4 slots per subframe. The slot duration is 0.25 ms, the subcarrier spacing is 60 kHz, and the symbol duration is approximately 16.67 μs.
(145) A resource grid may be used to represent the frame structure. Each time slot includes a resource block (RB) (also referred to as physical RBs (PRBs)) that extends 12 consecutive subcarriers. The resource grid is divided into multiple resource elements (REs). The number of bits carried by each RE depends on the modulation scheme.
(146) As illustrated in FIG. **3**A, some of the REs carry reference (pilot) signals (RS) for a UE (e.g., UE **104** of FIGS. **1** and **2**). The RS may include demodulation RS (DM-RS) (indicated as Rx for one particular configuration, where 100× is the port number, but other DM-RS configurations are

possible) and channel state information reference signals (CSI-RS) for channel estimation at the UE. The RS may also include beam measurement RS (BRS), beam refinement RS (BRRS), and phase tracking RS (PT-RS).
(147) FIG. **3**B illustrates an example of various DL channels within a subframe of a frame. The physical downlink control channel (PDCCH) carries DCI within one or more control channel elements (CCEs), each CCE including nine RE groups (REGs), each REG including four consecutive REs in an OFDM symbol.
(148) A primary synchronization signal (PSS) may be within symbol 2 of particular subframes of a frame. The PSS is used by a UE (e.g., **104** of FIGS. **1** and **2**) to determine subframe/symbol timing and a physical layer identity.
(149) A secondary synchronization signal (SSS) may be within symbol 4 of particular subframes of a frame. The SSS is used by a UE to determine a physical layer cell identity group number and radio frame timing.
(150) Based on the physical layer identity and the physical layer cell identity group number, the UE can determine a physical cell identifier (PCI). Based on the PCI, the UE can determine the locations of the aforementioned DM-RS. The physical broadcast channel (PBCH), which carries a master information block (MIB), may be logically grouped with the PSS and SSS to form a synchronization signal (SS)/PBCH block. The MIB provides a number of RBs in the system bandwidth and a system frame number (SFN). The physical downlink shared channel (PDSCH) carries user data, broadcast system information not transmitted through the PBCH such as system information blocks (SIB s), and paging messages.
(151) As illustrated in FIG. **3**C, some of the REs carry DM-RS (indicated as R for one particular configuration, but other DM-RS configurations are possible) for channel estimation at the base station. The UE may transmit DM-RS for the physical uplink control channel (PUCCH) and DM-RS for the physical uplink shared channel (PUSCH). The PUSCH DM-RS may be transmitted in the first one or two symbols of the PUSCH. The PUCCH DM-RS may be transmitted in different configurations depending on whether short or long PUCCHs are transmitted and depending on the particular PUCCH format used. The UE may transmit sounding reference signals (SRS). The SRS may be transmitted in the last symbol of a subframe. The SRS may have a comb structure, and a UE may transmit SRS on one of the combs. The SRS may be used by a base station for channel quality estimation to enable frequency-dependent scheduling on the UL.
(152) FIG. **3**D illustrates an example of various UL channels within a subframe of a frame. The PUCCH may be located as indicated in one configuration. The PUCCH carries uplink control information (UCI), such as scheduling requests, a channel quality indicator (CQI), a precoding matrix indicator (PMI), a rank indicator (RI), and HARQ ACK/NACK feedback. The PUSCH carries data, and may additionally be used to carry a buffer status report (BSR), a power headroom report (PHR), and/or UCI.
Additional Considerations
(153) The preceding description provides examples of sounding and transmission precoding matrix indication determination for uplink transmissions using machine learning models in communication systems. The preceding description is provided to enable any person skilled in the art to practice the various aspects described herein. The examples discussed herein are not limiting of the scope, applicability, or aspects set forth in the claims. Various modifications to these aspects will be readily apparent to those skilled in the art, and the generic principles defined herein may be applied to other aspects. For example, changes may be made in the function and arrangement of elements discussed without departing from the scope of the disclosure. Various examples may omit, substitute, or add various procedures or components as appropriate. For instance, the methods described may be performed in an order different from that described, and various steps may be added, omitted, or combined. Also, features described with respect to some examples may be combined in some other examples. For example, an apparatus may be implemented or a method

may be practiced using any number of the aspects set forth herein. In addition, the scope of the disclosure is intended to cover such an apparatus or method that is practiced using other structure, functionality, or structure and functionality in addition to, or other than, the various aspects of the disclosure set forth herein. It should be understood that any aspect of the disclosure disclosed herein may be embodied by one or more elements of a claim.

(154) The techniques described herein may be used for various wireless communication technologies, such as 5G (e.g., 5G NR), 3GPP Long Term Evolution (LTE), LTE-Advanced (LTE-A), code division multiple access (CDMA), time division multiple access (TDMA), frequency division multiple access (FDMA), orthogonal frequency division multiple access (OFDMA), single-carrier frequency division multiple access (SC-FDMA), time division synchronous code division multiple access (TD-SCDMA), and other networks. The terms "network" and "system" are often used interchangeably. A CDMA network may implement a radio technology such as Universal Terrestrial Radio Access (UTRA), cdma2000, and others. UTRA includes Wideband CDMA (WCDMA) and other variants of CDMA. cdma2000 covers IS-2000, IS-95 and IS-856 standards. A TDMA network may implement a radio technology such as Global System for Mobile Communications (GSM). An OFDMA network may implement a radio technology such as NR (e.g. 5G RA), Evolved UTRA (E-UTRA), Ultra Mobile Broadband (UMB), IEEE 802.11 (Wi-Fi), IEEE 802.16 (WiMAX), IEEE 802.20, Flash-OFDMA, and others. UTRA and E-UTRA are part of Universal Mobile Telecommunication System (UMTS). LTE and LTE-A are releases of UMTS that use E-UTRA. UTRA, E-UTRA, UMTS, LTE, LTE-A and GSM are described in documents from an organization named "3rd Generation Partnership Project" (3GPP). cdma2000 and UMB are described in documents from an organization named "3rd Generation Partnership Project 2" (3GPP2). NR is an emerging wireless communications technology under development.

(155) The various illustrative logical blocks, modules and circuits described in connection with the present disclosure may be implemented or performed with a general purpose processor, a DSP, an ASIC, a field programmable gate array (FPGA) or other programmable logic device (PLD), discrete gate or transistor logic, discrete hardware components, or any combination thereof designed to perform the functions described herein. A general-purpose processor may be a microprocessor, but in the alternative, the processor may be any commercially available processor, controller, microcontroller, or state machine. A processor may also be implemented as a combination of computing devices, e.g., a combination of a DSP and a microprocessor, a plurality of microprocessors, one or more microprocessors in conjunction with a DSP core, a system on a chip (SoC), or any other such configuration.

(156) If implemented in hardware, an example hardware configuration may comprise a processing system in a wireless node. The processing system may be implemented with a bus architecture. The bus may include any number of interconnecting buses and bridges depending on the specific application of the processing system and the overall design constraints. The bus may link together various circuits including a processor, machine-readable media, and a bus interface. The bus interface may be used to connect a network adapter, among other things, to the processing system via the bus. The network adapter may be used to implement the signal processing functions of the PHY layer. In the case of a user equipment (see FIG. **1**), a user interface (e.g., keypad, display, mouse, joystick, touchscreen, biometric sensor, proximity sensor, light emitting element, and others) may also be connected to the bus. The bus may also link various other circuits such as timing sources, peripherals, voltage regulators, power management circuits, and the like, which are well known in the art, and therefore, will not be described any further. The processor may be implemented with one or more general-purpose and/or special-purpose processors. Examples include microprocessors, microcontrollers, DSP processors, and other circuitry that can execute software. Those skilled in the art will recognize how best to implement the described functionality for the processing system depending on the particular application and the overall design constraints imposed on the overall system.

(157) If implemented in software, the functions may be stored or transmitted over as one or more instructions or code on a computer readable medium. Software shall be construed broadly to mean instructions, data, or any combination thereof, whether referred to as software, firmware, middleware, microcode, hardware description language, or otherwise. Computer-readable media include both computer storage media and communication media including any medium that facilitates transfer of a computer program from one place to another. The processor may be responsible for managing the bus and general processing, including the execution of software modules stored on the machine-readable storage media. A computer-readable storage medium may be coupled to a processor such that the processor can read information from, and write information to, the storage medium. In the alternative, the storage medium may be integral to the processor. By way of example, the machine-readable media may include a transmission line, a carrier wave modulated by data, and/or a computer readable storage medium with instructions stored thereon separate from the wireless node, all of which may be accessed by the processor through the bus interface. Alternatively, or in addition, the machine-readable media, or any portion thereof, may be integrated into the processor, such as the case may be with cache and/or general register files. Examples of machine-readable storage media may include, by way of example, RAM (Random Access Memory), flash memory, ROM (Read Only Memory), PROM (Programmable Read-Only Memory), EPROM (Erasable Programmable Read-Only Memory), EEPROM (Electrically Erasable Programmable Read-Only Memory), registers, magnetic disks, optical disks, hard drives, or any other suitable storage medium, or any combination thereof. The machine-readable media may be embodied in a computer-program product.
(158) A software module may comprise a single instruction, or many instructions, and may be distributed over several different code segments, among different programs, and across multiple storage media. The computer-readable media may comprise a number of software modules. The software modules include instructions that, when executed by an apparatus such as a processor, cause the processing system to perform various functions. The software modules may include a transmission module and a receiving module. Each software module may reside in a single storage device or be distributed across multiple storage devices. By way of example, a software module may be loaded into RAM from a hard drive when a triggering event occurs. During execution of the software module, the processor may load some of the instructions into cache to increase access speed. One or more cache lines may then be loaded into a general register file for execution by the processor. When referring to the functionality of a software module below, it will be understood that such functionality is implemented by the processor when executing instructions from that software module.
(159) As used herein, a phrase referring to “at least one of” a list of items refers to any combination of those items, including single members. As an example, “at least one of: a, b, or c” is intended to cover a, b, c, a-b, a-c, b-c, and a-b-c, as well as any combination with multiples of the same element (e.g., a-a, a-a-a, a-a-b, a-a-c, a-b-b, a-c-c, b-b, b-b-b, b-b-c, c-c, and c-c-c or any other ordering of a, b, and c).
(160) As used herein, the term “determining” encompasses a wide variety of actions. For example, “determining” may include calculating, computing, processing, deriving, investigating, looking up (e.g., looking up in a table, a database or another data structure), ascertaining and the like. Also, “determining” may include receiving (e.g., receiving information), accessing (e.g., accessing data in a memory) and the like. Also, “determining” may include resolving, selecting, choosing, establishing and the like.
(161) The methods disclosed herein comprise one or more steps or actions for achieving the methods. The method steps and/or actions may be interchanged with one another without departing from the scope of the claims. In other words, unless a specific order of steps or actions is specified, the order and/or use of specific steps and/or actions may be modified without departing from the scope of the claims. Further, the various operations of methods described above may be performed

by any suitable means capable of performing the corresponding functions. The means may include various hardware and/or software component(s) and/or module(s), including, but not limited to a circuit, an application specific integrated circuit (ASIC), or processor. Generally, where there are operations illustrated in figures, those operations may have corresponding counterpart means-plus-function components with similar numbering.

(162) The following claims are not intended to be limited to the aspects shown herein, but are to be accorded the full scope consistent with the language of the claims. Within a claim, reference to an element in the singular is not intended to mean “one and only one” unless specifically so stated, but rather “one or more.” Unless specifically stated otherwise, the term “some” refers to one or more. No claim element is to be construed under the provisions of 35 U.S.C. § 112(f) unless the element is expressly recited using the phrase “means for” or, in the case of a method claim, the element is recited using the phrase “step for.” All structural and functional equivalents to the elements of the various aspects described throughout this disclosure that are known or later come to be known to those of ordinary skill in the art are expressly incorporated herein by reference and are intended to be encompassed by the claims. Moreover, nothing disclosed herein is intended to be dedicated to the public regardless of whether such disclosure is explicitly recited in the claims.

Claims

1. A method for wireless communications by a user equipment (UE), comprising: generating a sounding reference signal (SRS) using an SRS deep neural network (DNNs); transmitting, to a network entity, the generated SRS in one or more resource elements (REs); receiving, from the network entity, information identifying a precoding matrix to use for uplink transmissions on a shared channel; precoding uplink transmissions based on the identified precoding matrix; transmitting, to the network entity, the precoded uplink transmissions on the shared channel; and receiving, from the network entity, a transmitted precoding matrix indicator (TPMI), based on the generated SRS, for use in precoding the uplink transmissions, wherein precoding the uplink transmissions comprises: applying a precoding to the uplink transmissions based, at least in part, on one or more precoding DNNs, given the TPMI as input.
2. The method of claim 1, wherein parameters of the SRS DNN comprise a set of weights and biases defined in a wireless communication standard.
3. The method of claim 1, further comprising receiving, from the network entity, configuration information identifying parameters of the SRS DNN.
4. The method of claim 3, wherein: the configuration information identifying parameters of the SRS DNNs comprises an index in a lookup table, and the index in the lookup table is mapped to a set of bias and weight parameters for the SRS DNN.
5. The method of claim 3, wherein the configuration information includes values for one or more bias and weight parameters of the SRS DNN.
6. The method of claim 1, wherein the generated SRS is transmitted on one or more statically defined resource elements.
7. The method of claim 1, further comprising receiving, from the network entity, configuration information identifying the one or more REs.
8. The method of claim 1, further comprising applying a code division multiplexing (CDM) cover code to the generated SRS prior to transmitting the generated SRS to the network entity.
9. The method of claim 1, wherein parameters of the one or more precoding DNNs comprise a statically defined set of weights and biases.
10. The method of claim 1, further comprising receiving, from the network entity, configuration information identifying parameters of the one or more precoding DNNs.
11. A method for wireless communications by a network entity, comprising: receiving, from a user equipment (UE), a sounding reference signal (SRS) generated by an SRS deep neural network

(DNN); identifying a precoding matrix based on a precoding matrix deep neural network and the received SRS; transmitting, to the UE, the information identifying the precoding matrix; receiving, from the UE, uplink transmissions precoded using the identified precoding matrix; and transmitting, to the UE, a transmitted precoding matrix indicator (TPMI), based on the generated SRS, for use in precoding the uplink transmissions via one or more precoding DNNs.
12. The method of claim 11, further comprising transmitting, to the UE, configuration information identifying parameters of the SRS DNN.
13. The method of claim 12, wherein: the configuration information identifying parameters of the SRS DNN comprises an index in a lookup table, and the index in the lookup table is mapped to a set of bias and weight parameters for the SRS DNN.
14. The method of claim 12, wherein the configuration information includes values for one or more bias and weight parameters of the SRS DNN.
15. The method of claim 11, further comprising transmitting, to the UE, configuration information identifying one or more REs on which the SRS is to be received.
16. The method of claim 11, further comprising transmitting, to the UE, configuration information identifying parameters of the one or more precoding DNNs.
17. The method of claim 11, further comprising transmitting, to the UE, configuration information identifying an association between a TPMI and a precoding to be applied to the uplink transmissions.
18. The method of claim 11, wherein the configuration information identifying the association between the received TPMI and the precoding information comprises a probabilistic model configured to identify a precoding to apply to the uplink transmissions based on a probability distribution over a universe of precodings, given the received TPMI as input.
19. An apparatus for wireless communication at a user equipment (UE), comprising: at least one memory comprising computer-executable instructions; and one or more processors configured to execute the computer-executable instructions and cause the UE to: generate a sounding reference signal (SRS) using an SRS deep neural network (DNNs); transmitting, to a network entity, the generated SRS in one or more resource elements (REs); receive, from the network entity, information identifying a precoding matrix to use for uplink transmissions on a shared channel; precode uplink transmissions based on the identified precoding matrix; transmit, to the network entity, the precoded uplink transmissions on the shared channel; and receiving, from the network entity, a transmitted precoding matrix indicator (TPMI), based on the generated SRS, for use in precoding the uplink transmissions, wherein precoding the uplink transmissions comprises: applying a precoding to the uplink transmissions based, at least in part, on one or more precoding DNNs, given the TPMI as input.
20. An apparatus for wireless communication at a network entity, comprising: at least one memory comprising computer-executable instructions; and one or more processors configured to execute the computer-executable instructions and cause the network entity to: receive, from a user equipment (UE), a sounding reference signal (SRS) generated by an SRS deep neural network (DNN); identify a precoding matrix based on a precoding matrix deep neural network and the received SRS; transmit, to the UE, the information identifying the precoding matrix; receive, from the UE, uplink transmissions precoded using the identified precoding matrix; and transmitting, to the UE, a transmitted precoding matrix indicator (TPMI), based on the generated SRS, for use in precoding the uplink transmissions via one or more precoding DNNs.